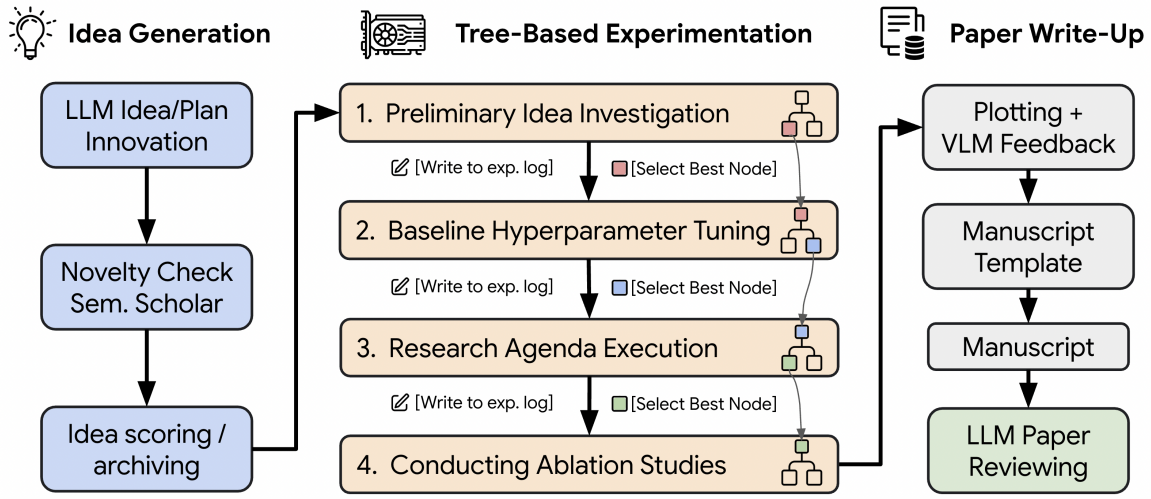

Figure 1 | **THE AI SCIENTIST-V2 Workflow.** The workflow consists of several phases covering automated idea generation, experiment execution, figure visualization, manuscript writing, and reviewing. Unlike the initial version, THE AI SCIENTIST-V2 removes the dependency on human-coded templates. Instead, it employs agentic tree search (managed by an Experiment Progress Manager across several stages, orange) to generate and refine code implementations. Subsequent experimentation leverages the best-performing code checkpoints (nodes) from the tree search to iteratively test various research hypotheses.

2. Background

THE AI SCIENTIST-V1 (Lu et al., 2024) introduced the first AI system that entirely automates scientific discovery and the presentation of its results. Given a baseline code template, it autonomously wrote code, executed experiments, visualized outcomes, and produced a complete scientific manuscript. However, despite representing a significant step forward, THE AI SCIENTIST-V1 was subject to limitations. Foremost among these was its reliance on human-crafted baseline code templates, significantly constraining its autonomy and hindering unconstrained out-of-the-box deployability. Instead, human effort was still required to draft an initial base experiment outline in code. Additionally, the experimentation process followed a strictly linear hypothesis-testing routine, limiting depth and exploration flexibility, especially when addressing complex research questions.

Language Model Agent Scaffolding. To further enhance LLM performance on complex reasoning tasks, researchers have developed agentic scaffolding frameworks, each with distinct advantages and limitations. For example, Reflexion (Shinn et al., 2024) enables models to iteratively reflect on previous responses, encouraging self-improvement through critical evaluation of past outputs; it improves robustness, but can introduce computational overhead and slower inference. Another promising direction is the integration of tree-search strategies with LLMs (Jiang et al., 2025), allowing structured exploration of reasoning paths. This approach enhances systematic reasoning and comprehensiveness, though at the cost of increased complexity, higher computational demands, and challenges in scalability.

Tree Search with Large Language Models. We empirically observed that automated research conducted by THE AI SCIENTIST-V1 often resulted in short-sighted experimentation. The human-driven scientific process, on the other hand, relies on open-ended hypothesis generation, stepping-stone collection, and iterative hypothesis refinement. Recent advances using code generation as an action space have opened new opportunities for LLM-driven automated workflows (Wang et al., 2024). AIDE (Jiang et al., 2025) combines LLM-based code generation with tree search, demon-